13.14 Solution: (a) Following the discussion in Section 13.7, the radiated field contribution from one medium, either the foil ($m = 0$) or the air gap ($m = 1$),can be written as

$$\mathbf{E}_{\mathrm{rad,m}} = \frac{e^{ikr}}{r}\left(-\frac{\omega_p^2}{4\pi c^2}\right)\int_{z'_{\mathrm{min}}}^{z'_{\mathrm{max}}} \rho_m(\hat{\mathbf{k}}\times\mathbf{E}_i)\times\hat{\mathbf{k}}\exp\left(-i\mathbf{k}_m\cdot\mathbf{x}'\right)d^3x'.$$

Here, we have exploited the fact that ρ and $\mathbf{k}$ are actually constant in one medium. Also, for the foil, $\rho = \rho(0) \equiv 1$, and for air gap, $\rho \equiv 0$. Then, we can formally write the radiate field as

$$\mathbf{E}_{\mathrm{rad}} = \frac{e^{ikr}}{r}\left(-\frac{\omega_p^2}{4\pi c^2}\right)\int_{z'>0} \rho(z')(\hat{\mathbf{k}}\times\mathbf{E}_i)\times\hat{\mathbf{k}}\exp\left(-i\cos\theta\int^{z} k(z')dz'\right)\exp\left(-i\right)d^3x'.$$